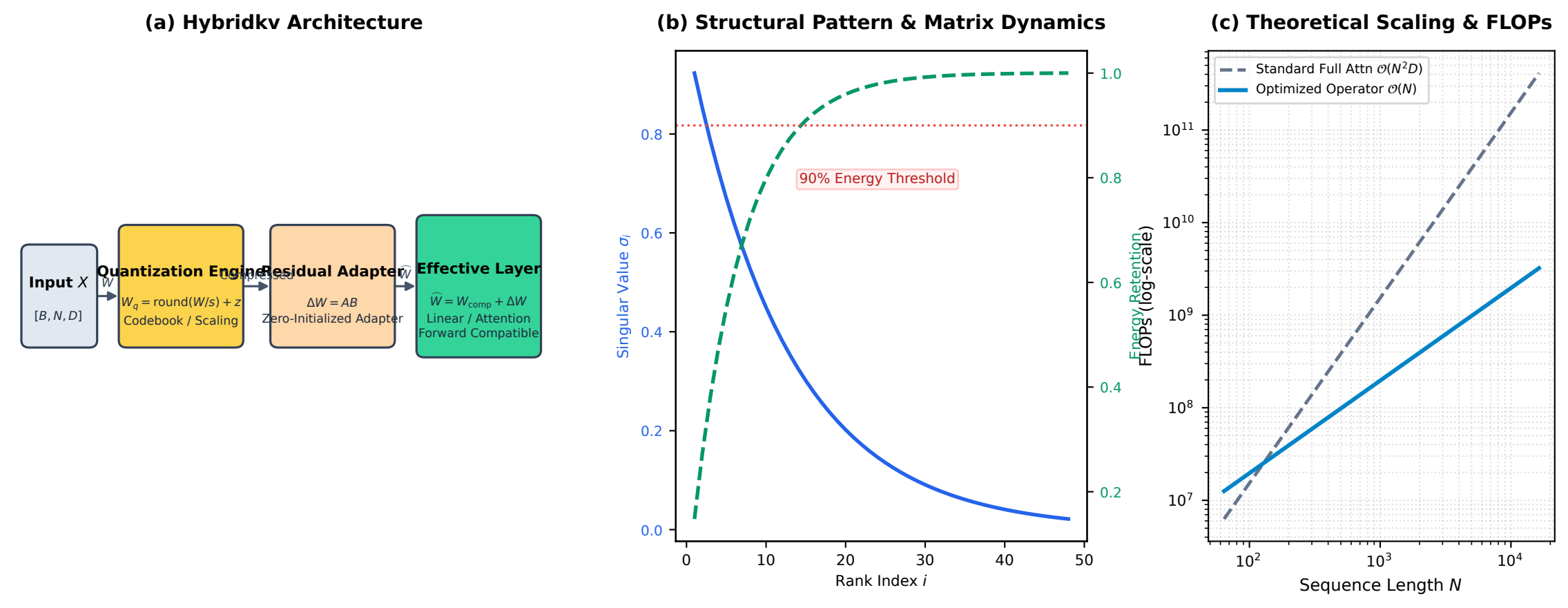
(a) Hybridkv Architecture
Input $X$
$[B, N, D]$
$W$
Quantization Engine
$W_q = \text{round}(W/s) + z$
Codebook / Scaling
Compressed
Residual Adapter
$\Delta W = AB$
Zero-Initialized Adapter
$\widehat{W}$
Effective Layer
$\widehat{W} = W_{\text{comp}} + \Delta W$
Linear / Attention
Forward Compatible
(b) Structural Pattern & Matrix Dynamics
Singular Value $\sigma_i$
Energy Retention
90% Energy Threshold
Rank Index $i$
(c) Theoretical Scaling & FLOPs
Standard Full Attn $\mathcal{O}(N^2D)$
Optimized Operator $\mathcal{O}(N)$
FLOPs (log-scale)
Sequence Length $N$